

ForgeNet-X: A Rule-Based Computer Vision Framework for Handwriting Synthesis and Forensic Document Authentication

Hariom Sharma

Department of Computer Science and Engineering

[Institution Name], [City], [Country]

[author.email@institution.edu]

Abstract—Forensic document analysis and synthetic handwriting generation present significant challenges in digital security and authentication research. Contemporary solutions predominantly employ neural network architectures that demand large annotated corpora and substantial computational infrastructure. This paper presents ForgeNet-X, a classical computer vision framework delivering five integrated capabilities—handwriting preprocessing, character-level segmentation, handwriting synthesis, signature forgery detection, and image provenance classification—entirely without trained machine learning models. The segmentation pipeline employs connected component analysis (CCA), histogram-valley line detection, seam-carving for ambiguous character separation, polynomial baseline fitting, and satellite-merge correction. Handwriting synthesis leverages a Times New Roman proportional normalisation system, blending author-measured typographic ratios with reference proportions and aligning characters to a shared baseline via exponential smoothing. Signature verification computes a four-metric composite score aggregating SSIM, Hu moment matching, histogram correlation, and pixel MSE to classify signatures as Low, Moderate, or High Risk. Provenance detection applies five EXIF and pixel-level heuristics to determine whether a document image originates from a physical imaging device or a digital generation pipeline. Experiments on diverse handwriting samples demonstrate accurate segmentation and reliable signature discrimination, with the complete system deployable on commodity hardware without GPU acceleration.

Index Terms—character segmentation, connected component analysis, document forensics, handwriting synthesis, image provenance, signature verification, SSIM.

I. INTRODUCTION

Digital forensics increasingly demands tools capable of analysing handwritten document authenticity, detecting signature forgeries, and identifying the provenance of scanned or digitally generated images. The proliferation of consumer-grade image-editing software has lowered the barrier for document manipulation, placing pressure on analysts who must determine document origin under legal scrutiny that requires transparent, auditable decisions.

Existing automatic approaches rely predominantly on neural networks: convolutional architectures for feature extraction, recurrent networks for sequence modelling, and generative adversarial networks for synthesis. While powerful, these models require large labelled datasets, GPU acceleration, and domain-specific training, making them poorly suited to resource-constrained or legally sensitive environments where interpretability is paramount.

ForgeNet-X addresses this gap with a fully classical computer vision pipeline comprising five interdependent modules: (1) adaptive preprocessing via CLAHE and Otsu binarisation; (2) character segmentation using CCA with seam-carving disambiguation; (3) rule-based handwriting

synthesis calibrated by author-specific typographic measurements; (4) four-metric composite signature verification; and (5) heuristic image provenance scoring. The system is deployed as a Flask web application, generates annotated forensic PDF reports, and requires no model training.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the system architecture. Section IV details each methodology component. Section V presents experimental results. Section VI concludes with directions for future work.

II. RELATED WORK

Handwriting analysis has been extensively studied since the foundational taxonomy established by Plamondon and Srihari [1], which distinguishes offline (image-based) from online (trajectory-based) approaches. Offline segmentation methods range from projection-profile histograms [2] to graph-cut techniques [3] and deep segmentation networks [4]. Seam-carving, originally proposed for content-aware image resizing by Avidan and Shamir [5], has been adapted for character separation in connected handwriting scripts.

Signature verification has evolved from simple threshold comparisons to composite feature ensembles. Hafemann et al. [6] demonstrated deep CNN-based verification achieving high accuracy on CEDAR and GPDS datasets, while Impedovo and Pirlo [7] note that classical feature-based methods remain competitive for skilled forgery scenarios. The Structural Similarity Index Measure (SSIM), introduced by Wang et al. [13], finds broad application in signature and image quality assessment owing to its perceptual alignment with the human visual system.

Image provenance and manipulation detection has gained urgency in the era of GAN-generated imagery. Farid [8] surveys classical heuristics including EXIF metadata analysis, noise inconsistency, and compression artifact detection. Zhang et al. [9] apply CNNs to detect GAN-specific artifacts; however, classical heuristics retain value where independence from training data and decision interpretability are required.

ForgeNet-X distinguishes itself from prior work by unifying all five forensic capabilities into a single training-free framework with fully transparent, auditable decision logic.

III. SYSTEM ARCHITECTURE

ForgeNet-X is implemented as a modular Python 3.11 application served by Flask 2.x. Five utility modules—preprocessing, segmentation, normalisation, signature analysis, and provenance detection—are orchestrated by a central application layer that manages HTTP routing, file storage, audit logging, and PDF report generation.

The upload flow for a handwriting sample proceeds as follows. The uploaded image is passed through the preprocessing module to yield a normalised binary ink mask and five intermediate pipeline visualisations. The segmentation module analyses the binary image and produces a JSON character manifest recording bounding boxes, ink-region sub-bounds, line indices, and baseline offsets. The normalisation module consumes the manifest to construct a baseline-aligned character atlas. The synthesis module renders new text from the atlas using exponentially smoothed baseline placement.

Signature analysis and provenance detection operate as independent pipelines: the signature module accepts two image paths and returns a composite similarity score with per-metric breakdown and style features, while the provenance module accepts one image path and returns a synthetic-origin score with supporting flag evidence. Results from all modules are consolidated into a structured PDF forensic report via a ReportLab-based generator. Generation events are appended to a JSON Lines audit file for traceability and compliance.

A. Technology Stack

The implementation uses OpenCV 4.x for image processing, NumPy and SciPy for numerical computation, scikit-image for SSIM computation, Pillow for EXIF metadata extraction, and ReportLab for PDF generation. No deep learning framework is required. The application runs on standard CPU hardware, with total package requirements under 200 MB.

B. Web Interface and Reporting

A responsive single-page front end communicates with the Flask back end via REST API calls. The interface exposes three workflows: handwriting upload and synthesis, signature pair comparison, and combined forensic report download. The PDF report includes preprocessing pipeline visualisations, a segmentation overview, the character atlas sheet, signature comparison metrics, and provenance analysis results with confidence indicators colour-coded green (Low Risk), orange (Moderate), or red (High Risk).

IV. METHODOLOGY

A. Preprocessing Pipeline

Preprocessing matters more than it might seem. Handwriting scans routinely suffer from uneven illumination, paper grain, scanner noise, and highly variable pen pressure—problems that will confuse even the most carefully designed segmentation logic if they are not addressed first.

We start with a grayscale conversion, then immediately apply CLAHE [10] with a clip limit of 2.0 and an 8×8 tile grid. The region-aware nature of CLAHE is the key advantage: a shadow in one corner of the page should not skew the binarisation threshold for the whole document, which is precisely the failure mode of global histogram equalisation. Otsu's method [11] then selects the threshold that best separates the ink and background pixel distributions—no manual tuning required. A 2×2 morphological opening clears any salt-and-pepper noise that survives binarisation, and the mask is inverted to the black-ink-on-white convention the rest of the pipeline expects.

B. Character Segmentation

Separating individual characters from a handwriting image is, in our experience, where most classical pipelines fall apart. The challenge is not finding the ink—it is deciding where one character ends and the next begins, especially in cursive or overlapping scripts.

We tackle this in two passes: first split the image into text lines, then split each line into characters. Line boundaries come from horizontal projection profiles—we sum ink pixels row by row and look for valleys, stretches of low-density rows that correspond to inter-line white space. Because handwriting rarely sits flat on an invisible baseline, we fit a polynomial curve to the centroid y-coordinates of the connected components within each candidate line [12]. This gracefully absorbs moderate drift and the vertical intrusion of ascenders and descenders from adjacent lines.

Within each line, 8-connectivity CCA gives us a set of candidate blobs. Small blobs—letter dots, accent marks, ink specks—are not discarded; they are merged back into their nearest spatial neighbour, because throwing away ink is worse than misattributing it. The genuinely hard case is two characters whose bounding boxes overlap horizontally. For those, we apply seam carving [5]: a dynamic programming pass finds the lowest-cost vertical cut between the two regions, threading through white space rather than ink, and uses it as the split boundary. It is a clean, principled solution that avoids any need for character-shape heuristics. Every accepted region is then saved with a manifest entry recording

its label, bounding box, ink-region sub-bounds, line index, and baseline offset.

C. Handwriting Synthesis and Normalisation

Synthesising text that actually looks like a particular person requires answering a deceptively simple question: how tall should each character be? A capital A from one writer may be twice the height of a lowercase a; another writer may keep them nearly the same. Flattening everything to one pixel height strips away that personal typographic signature entirely.

We begin by reading the character manifest and measuring five author-specific category heights: x-height (short lowercase letters), ascender height, descender depth, cap height, and digit height. Together these five numbers define the writer's proportional system. When a category is absent from the sample—say the writer used no uppercase letters—we fall back to inferring the missing height from the ones we did measure, using fixed Times New Roman typographic ratios as a reference.

The target height for each character category n is a weighted blend of the author's own measured height h^{na} and the TNR reference height h^{nT} , controlled by a single dial $\beta \in [0, 1]$:

$$h^n = (1 - \beta) \cdot h^{na} + \beta \cdot h^{nT} \quad (1)$$

Setting $\beta = 0$ reproduces the author's style exactly, idiosyncrasies and all. Setting $\beta = 1$ produces perfectly proportioned TNR geometry. We recommend $\beta = 0.35$: it quietly corrects the most jarring size inconsistencies while leaving the personal character of the handwriting intact. Each normalised character image is embedded in a canvas tall enough to hold the full ascender-to-descender range and annotated with a `baseline_y` value—the pixel row at which the typographic baseline falls—so that different-sized characters can be vertically aligned precisely during rendering.

Real handwriting on a ruled surface is not perfectly flat; the baseline wanders slightly from character to character. We model this with an exponential smoother that keeps the drift bounded and visually plausible. At each character position i the vertical offset d_i evolves as:

$$d_i = \alpha \cdot t_i + (1 - \alpha) \cdot d_{i-1}, \quad d_i \leftarrow 0.96 \cdot d_i \quad (2)$$

Here $\alpha = 0.18$ controls how quickly the offset responds to a new drift target t_i , which is resampled randomly with probability 0.25 at each step. The factor 0.96 pulls the drift back toward zero, preventing it from wandering to one side of the page. The result looks like someone writing carefully on a ruled surface rather than a perfectly mechanical straight line or a scribble slanting downhill.

D. Signature Forgery Analysis

No single image similarity metric tells the whole story of a signature comparison. Pixel-wise difference misses structural similarity; shape moments miss fine texture; SSIM alone can be fooled by a signature that is the right shape but the wrong ink density. We therefore compute four independent metrics and combine them into a single composite score $S \in [0, 100]$:

$$S = 0.40 \cdot S_{\text{SSIM}} + 0.25 \cdot S_{\text{Hu}} + 0.20 \cdot S_{\text{hist}} + 0.15 \cdot S_{\text{pixel}} \quad (3)$$

Both images are binarised (Gaussian blur followed by Otsu inversion) and rescaled to 256×256 before any metric is

computed. S_{SSIM} is the Structural Similarity Index [13] scaled to $[0, 100]$; it captures luminance, contrast, and local structure simultaneously and earns the highest weight (0.40) because skilled forgeries tend to reproduce the gross shape of a signature but betray themselves in exactly those three dimensions. S_{Hu} uses the seven Hu moment invariants [14], log-transformed for numerical stability, mapped to $[0, 100]$ via $100 \cdot \exp(-0.1d)$ where d is the Euclidean distance between log-moment vectors. S_{hist} is the normalised grayscale histogram correlation remapped from $[-1, 1]$ to $[0, 100]$. $S_{\text{pixel}} = 100 \cdot (1 - \text{MSE}/65025)$ penalises raw pixel-level disagreement.

A composite score of 80 or above is classified as Low Risk; 50–79 as Moderate Risk; below 50 as High Risk. We surface the individual subscores alongside extracted style features—ink density, aspect ratio, estimated stroke width, centroid position, and stroke-segment count—so a human analyst can understand exactly which dimension of the comparison raised the alarm, rather than receiving just a single opaque verdict.

E. Image Provenance Detection

Determining whether a document image came from a real scanner or was generated by software is fundamentally a question of looking for things that cameras leave behind and rendering engines do not—or vice versa. We translate that intuition into five concrete, independently scored heuristic checks that accumulate a synthetic-origin score $P \in [0, 100]$:

(1) No EXIF metadata (+15 pts). Every camera and most scanner firmwares write EXIF headers. A completely EXIF-free image is therefore suspicious—though not conclusive, since some export workflows strip metadata intentionally.

(2) EXIF Software tag matches a known digital tool (+40 pts). This is the single most reliable signal, which is why it carries the highest weight. The Software field often names the exact application—Photoshop, Pillow, OpenCV, GIMP, Stable Diffusion, and so on—leaving little room for ambiguity. When it fires, the image is almost certainly not a genuine scan.

(3) No camera make or model (+10 pts). Absent hardware identifiers are a softer signal, but they corroborate the other checks when present.

(4) Background more than 70 % pure white (+20 pts). A genuine handwriting scan picks up paper grain, scanner lamp non-uniformity, and shadow near the spine. A rendered image usually has a background that is pixel-perfectly white (value 255). This distinction is surprisingly reliable in practice.

(5) Background Laplacian variance below 8.0 (+15 pts). Even after the background-purity check passes, the texture of the background reveals its origin. Camera noise and paper grain produce high Laplacian variance; rendered backgrounds are almost completely flat. A variance under 8.0 is, in our testing, essentially never seen in real scans.

Images scoring below 30 are marked Likely Human; 30–59 Possibly Synthetic; 60 and above Likely Synthetic. Every check that fired is returned alongside the score, giving the analyst a clear trail of evidence rather than just a verdict.

V. RESULTS AND DISCUSSION

Experiments were conducted on a corpus of handwriting samples from five writers spanning print, cursive, and mixed styles, each scanned at 300 DPI. Signature analysis used 40 image pairs; provenance detection used 30 images. All tests were performed on a standard consumer laptop (Intel Core i5, 8 GB RAM) without GPU acceleration.

A. Segmentation Performance

Character segmentation achieved a correct-isolation rate of approximately 87 % across the test corpus, where correctness is defined as a bounding box containing exactly one complete character glyph. The primary failure mode was tight ligature connections in cursive script, where seam-carving occasionally threaded a path through a thin ink stroke rather than the true inter-character gap. Satellite-merge correctly attributed isolated ink fragments—letter dots and accent marks—to their parent characters in 93 % of cases, reducing the false-component rate significantly compared to unmerged CCA.

B. Synthesis Quality

Synthesised text was evaluated by three independent reviewers using a five-point legibility-and-naturalness scale. At the recommended blend factor $\beta = 0.35$, outputs averaged 4.1 / 5 for legibility and 3.8 / 5 for naturalness. Values below $\beta = 0.20$ preserved more author idiosyncrasy but occasionally produced height inconsistencies across character categories. Values above $\beta = 0.60$ produced visually uniform but stylistically generic text. All reviewers rated the exponential-smoothing baseline model as significantly more natural than independent per-character jitter.

C. Signature Verification

On 20 genuine-match pairs and 20 skilled-forgery pairs, the composite metric achieved a correct-classification rate of 82.5 % at default thresholds. SSIM contributed the most discriminative power for skilled forgeries, where global structure differs subtly. Hu moments were more robust to legitimate intra-signer variation across sessions. Ablation experiments showed a 4–8 % accuracy drop when any single metric was removed, confirming that all four components contribute complementary discriminative information.

D. Provenance Detection

All 15 digitally generated images scored ≥ 60 (Likely Synthetic). Among 15 genuine scanned documents, 13 scored below 30 (Likely Human) and 2 scored between 30 and 50 (Possibly Synthetic) owing to unusually clean scanning conditions that triggered the background-purity heuristic. No genuine scanned document was misclassified as Likely Synthetic, yielding zero false positives at the highest suspicion tier—critical in forensic contexts where falsely accusing a genuine document carries significant legal consequence.

VI. CONCLUSION

This paper presented ForgeNet-X, a rule-based computer vision framework for forensic handwriting analysis that integrates preprocessing, character segmentation, handwriting

synthesis, signature verification, and image provenance detection without any trained machine learning model. The framework achieves competitive performance across all five tasks on commodity CPU hardware, with transparent and auditable decision logic suitable for forensic, educational, and compliance-sensitive applications.

The primary limitation is segmentation sensitivity to highly cursive or overlapping handwriting styles. Future work will focus on four areas: (1) adaptive segmentation parameter tuning guided by writer-specific profiles; (2) GAN or diffusion-based synthesis for higher-fidelity stylistic output; (3) dynamic stroke-order signature analysis for improved cross-domain generalisation; and (4) multi-writer profile storage for comparative forensic queries across document collections. ForgeNet-X represents a transparent, interpretable complement to black-box deep learning systems and a practical baseline against which future learned approaches can be benchmarked.

References

- [1] R. Plamondon and S. N. Srihari, "Online and off-line handwriting recognition: A comprehensive survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 1, pp. 63–84, Jan. 2000.
- [2] C.-L. Liu, S. Jaeger, and M. Nakagawa, "Online recognition of Chinese characters: The state-of-the-art," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 2, pp. 198–213, Feb. 2004.
- [3] V. Lladós, E. Valveny, J. Lladós, and E. Martí, "Symbol recognition by error-tolerant subgraph matching between region adjacency graphs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 23, no. 10, pp. 1137–1143, Oct. 2001.
- [4] A. Graves, M. Liwicki, S. Fernández, R. Bertolami, H. Bunke, and J. Schmidhuber, "A novel connectionist system for unconstrained handwriting recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 5, pp. 855–868, May 2009.
- [5] S. Avidan and A. Shamir, "Seam carving for content-aware image resizing," *ACM Trans. Graph.*, vol. 26, no. 3, Art. 10, Jul. 2007.
- [6] L. G. Hafemann, R. Sabourin, and L. S. Oliveira, "Learning features for offline handwritten signature verification using deep convolutional neural networks," *Pattern Recognit.*, vol. 70, pp. 163–176, Oct. 2017.
- [7] D. Impedovo and G. Pirlo, "Automatic signature verification: The state of the art," *IEEE Trans. Syst. Man Cybern. C*, vol. 38, no. 5, pp. 609–635, Sep. 2008.
- [8] H. Farid, "Image forgery detection," *IEEE Signal Process. Mag.*, vol. 26, no. 2, pp. 16–25, Mar. 2009.
- [9] X. Zhang, S. Karaman, and S.-F. Chang, "Detecting and simulating artifacts in GAN fake images," in *Proc. IEEE Int. Workshop Inf. Forensics Secur. (WIFS)*, 2019, pp. 1–6.
- [10] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA, USA: Academic Press, 1994, pp. 474–485.
- [11] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst. Man Cybern.*, vol. 9, no. 1, pp. 62–66, Jan. 1979.
- [12] A. Vinciarelli, "A survey on off-line cursive word recognition," *Pattern Recognit.*, vol. 35, no. 7, pp. 1433–1446, Jul. 2002.
- [13] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [14] M.-K. Hu, "Visual pattern recognition by moment invariants," *IRE Trans. Inf. Theory*, vol. 8, no. 2, pp. 179–187, Feb. 1962.